

2025 CpSc H Q5

Section: Computer Systems

Topic: Computer Architecture and the Fetch-Execute Cycle

Question Summary

- (a) State how the processor uses the control bus during the fetch-execute cycle.
- (b)(i) State how a manufacturer could improve performance when designing a new processor.
- (b)(ii) Explain why this improves performance.

Worked Solution

(a) The control bus carries control signals (such as Read, Write, Clock, and Interrupt) that coordinate and manage the actions of other parts of the processor and memory during each stage of the cycle.

During the fetch stage, for example, the processor sends a Memory Read signal via the control bus to fetch the instruction from memory into the instruction register.

(b)(i) & (ii) Performance improvements and their explanations:

- Increase clock speed: Each instruction cycle completes faster. A higher clock frequency means more instructions per second.

→ Why: Each fetch-decode-execute sequence takes less time, so overall throughput increases.

- Add more processor cores: Each core can execute instructions independently.

→ Why: Multiple programs (or threads of the same program) can run in parallel, improving performance in multi-tasking or parallel workloads.

- Increase cache memory size: Cache stores recently used data close to the CPU.

→ Why: Accessing data from cache is much faster than from main memory, reducing waiting time and improving efficiency.

Final Answer

(a) The control bus sends timing and control signals such as Read and Write to coordinate the fetch-execute cycle.

(b)(i) Increase clock speed / add more cores / increase cache size.

(b)(ii) Faster clock = more cycles per second; more cores = parallel execution; larger cache = reduced memory access delays.

Revision Tips

- Know the three main buses: Address (where), Data (what), Control (how/when).
- Remember that control signals are one-way instructions (e.g. Read, Write, Clock, Interrupt).
- List multiple valid performance improvements, but link each one to why performance increases.
- SQA marking usually gives 1 mark for identifying and 1 for explaining each improvement.

